

AI Concepts for Official Statisticians

Executive decision brief

OS-A02-A02 | Official Statistics and AI Pathway | Version 1.0 | 26 September 2026

Leadership conclusion

AI may support official-statistics tasks when the statistical purpose, mechanism, evidence, failure consequences, controls and decision authority are explicit. Capability alone is not approval. Leaders should authorize bounded use only when performance is demonstrated for the intended population and production conditions, confidentiality and professional independence are protected, and named officials retain stopping and release authority.

Decision field	Required leadership position
Primary question	Which defined statistical task may be assisted, under what evidence and authority conditions?
Permitted outcomes	Permit a bounded experiment, redesign, reject, or escalate.
Non-compensatory gates	Mandate, confidentiality, statistical validity, professional independence and release authority.
Executive obligation	Fund evidence, reviewer capacity, monitoring, fallback and institutional learning - not software acquisition alone.
Canonical source	OS-A02-A01 AI Concepts for Official Statisticians v0.2, accepted 26 September 2026.

Immediate leadership actions

- Require every proposal to name the official product, user, target population, consequence of error and excluded decisions.
- Demand comparison with the current non-AI or simpler baseline using end-to-end quality and quality-adjusted time.
- Separate technical capability from authority to produce, revise or release official statistics.
- Stop any proposal that uses protected information in an unapproved service or treats fluent output as verified evidence.

What AI is and what it is not

For governance, an AI system is best described through observable properties: its inputs, objectives, inferred or encoded function, outputs, degree of autonomy, adaptation and operating conditions. Terms such as understands, knows or decides should be replaced by testable claims. The definition does not confer truthfulness, accountability, statistical mandate or professional judgement (OECD, 2024).

Mechanism	What it does	Leadership caution
Rules and workflow automation	Executes declared conditions and procedures.	Can reproduce an invalid specification perfectly.
Statistical and machine learning	Estimates relationships or predictive functions from data.	Predictive accuracy does not establish population validity or unbiased official estimates.
Deep learning	Learns layered representations from large or complex data.	Scale may improve fit while domain transfer and subgroup assurance remain weak.
Generative AI	Produces content conditionally from learned distributions.	Fluency and citation appearance do not verify factual or statistical validity.
Retrieval augmented systems	Selects sources and generates responses from retrieved context.	A cited answer may omit the controlling or current source.
Optimization and agents	Selects or executes actions using objectives, tools and feedback.	Technical autonomy cannot acquire institutional or release authority.

The governing distinction

AI may retrieve, rank, classify, predict, generate, translate, summarize, draft code and signal anomalies. The subject-matter statistician must define the construct, population, authoritative evidence, tolerances, uncertainty, interpretation, revision policy and release decision. IT, security, legal and procurement functions control infrastructure and supplier boundaries; they do not replace statistical authority.

Why this matters

Official statistics is a sociotechnical evidence system. A component error can propagate through frames, classifications, linkage, estimation, disclosure control and communication. Leaders must therefore govern the released claim and its evidence lineage, not merely a model score or vendor demonstration (United Nations, 2014; UNSD, 2019).

Seven questions before any AI decision

The following sequence is mandatory because it prevents a technology label from replacing statistical reasoning.

Step	Leadership question	Required evidence
1 Purpose	What public or statistical purpose is served, and what is the consequence of error?	Mandate, product, user, population, intended and excluded uses.
2 Work	Which tasks and professional judgements exist now?	Workflow map, baseline quality, effort, exceptions and handoffs.
3 Mechanism	What is encoded, estimated, retrieved, generated or optimized?	Inputs, objective, output, dependencies, adaptation and constraints.
4 Evidence	What would justify use in the intended setting?	Reference standard, evaluation population, metrics, uncertainty and tolerances.
5 Failure	What can fail, who is affected and how does error propagate?	Failure register, subgroup/domain analysis and product-level consequences.
6 Control	Who verifies, stops, corrects, falls back and escalates?	Named roles, capacity, access, logs, triggers and restart conditions.
7 Decision	Should the institution permit, redesign, reject or escalate?	Conditional record, residual uncertainty, monitoring and review date.

Decision vocabulary

Permit a bounded experiment only when the task, data, environment and authority are constrained. Redesign when value is plausible but evidence or controls are inadequate. Reject when mandatory gates fail or residual risk is disproportionate. Escalate when law, rights, public value or release authority exceeds the operating team's mandate.

A defensible rejection or redesign is a successful professional outcome. Productivity gains cannot compensate for a failed confidentiality, validity or authority gate.

Evidence thresholds and non negotiable gates

Evidence intensity should rise with consequence, uncertainty, irreversibility and scale. Evaluation must address the intended population and production workflow, not only a convenient development sample. Human review is itself a capacity-constrained process whose errors and costs must be measured (NIST, 2023; UNECE, 2025).

Evidence domain	Minimum executive requirement	Stop or redesign signal
Baseline	Current method measured using comparable quality, effort and error definitions.	Only vendor benchmark or no credible comparator.
Validity	Independent reference evidence, uncertainty and pre-declared tolerances.	Training fit, aggregate accuracy or anecdotal success presented as validation.
Population and subgroups	Domain, language, geography and affected-group performance with uncertainty.	Important groups absent, unstable or materially worse.
Workflow	End-to-end testing including verification, correction, exceptions and fallback.	Reviewer burden erases benefit or errors compound downstream.
Confidentiality and security	Approved data boundary, minimization, access control, retention and incident response.	Protected data enters an unapproved service or supplier boundary is unknown.
Reproducibility	Versions, data lineage, prompts or configurations, sources, outputs and approvals recorded.	Undisclosed model or corpus change; result cannot be reconstructed.
Authority	Named decision owner, reviewer, stopping power and release official.	System or supplier is treated as the accountable decision maker.

Quality adjusted time

Measure total elapsed and staff time for preparation, generation, verification, correction, exception handling, documentation and recovery. Faster generation with greater checking burden is not an efficiency gain. Comparison must also include continuity, supplier dependence and retained institutional capability.

Uncertainty

Model confidence, token probability and fluent language are not interchangeable with the probability that an official claim is correct. Leaders should require an uncertainty ledger identifying sampling, coverage, measurement, linkage, model, retrieval, generative and decision uncertainty rather than one reassuring score.

Five Kintaba decisions

The synthetic Kintaba cases show why one institution may reach different decisions for different tasks even when the same technology is available.

Case	Synthetic comparison	Decision boundary
Occupation coding	Manual baseline: 91.2% adjudicated accuracy. Candidate: 94.6% overall, but 81.3% recall for informal work and 76.8% for the smallest language strata.	REDESIGN. Repair labels and subgroup performance; permit suggestions only with adjudication.
Record linkage	Probabilistic baseline: 1.6% false links and 7.4% missed links. Learned matcher: 2.3% false links, 4.1% missed links, but 5.9% false links for recent migrants.	PERMIT bounded sandbox analysis with sensitivity analysis; REJECT operational register updating.
Crop area	Deep model raises pixel accuracy from 84.1% to 92.7%, yet understates area by 11.8% in intercropped zones and 15.4% for fragmented plots.	REDESIGN around calibrated estimation and ground surveys; exclude unsupported domains.
Release drafting	Generation takes 11 minutes plus 35 minutes verification versus 52 minutes analyst baseline; material corrections rise from 0.7 to 1.9 per release.	REDESIGN. Public inputs and claim traceability only; no autonomous interpretation or release.
Public query support	Candidate retrieves the controlling source in 93.8% overall but 82.0% for minority-language queries; correct refusal is 88.6% against 98% tolerance.	REDESIGN. Improve multilingual retrieval and refusal; retain answer, qualify, refuse or refer routes.

Executive lesson

Aggregate performance can conceal the precise domain in which an official-statistics use fails. A model may improve a component metric while worsening an estimator, excluding a language group, increasing false links, creating unsupported claims or shifting unmanageable work to reviewers. The decision must follow the use and consequence, not the novelty of the mechanism.

Institutional responsibilities

Meaningful human oversight requires competence, information, time, independence and power to intervene. A person who can only approve an opaque result is not exercising control.

Role	Non delegable responsibility
Senior statistical leadership	Set mandate, risk appetite and non-compensatory gates; resource evidence and control; designate release authority.
Subject-matter statisticians	Define constructs, classifications, populations, authoritative sources, tolerances, interpretation and product consequences.
Methodologists	Design evaluation, weighting, uncertainty, estimator integration, subgroup analysis and monitoring.
Data stewards and source owners	Control provenance, access, representativeness, linkage, retention and source-system change.
IT and cybersecurity	Control architecture, identity, permissions, logging, resilience, testing and incident containment.
Legal and procurement	Translate mandate, confidentiality, intellectual-property, audit, supplier-change and exit requirements into enforceable terms.
Reviewers and operators	Verify evidence, record corrections and exceptions, invoke stopping rules and maintain fallback competence.
Release official	Determine whether the official product may be released; authority never transfers to a system or supplier.

Supplier assurance questions

- What exactly changes after deployment: parameters, prompts, retrieval corpus, tools, thresholds or hosting?
- Which populations, languages, domains and failure conditions were not evaluated?
- Can the institution reconstruct outputs and receive advance notice of material change?
- What data, prompts, embeddings, logs and derived artefacts leave the institutional boundary?
- How will service continuity, model withdrawal, export, deletion and independent audit work?

Leadership implementation plan

A proportionate 90-day programme can convert policy intentions into testable institutional practice without authorizing production deployment.

Period	Leadership action	Evidence produced
Days 1 to 15	Select one approved bounded task; appoint statistical, methodological, data, security and authority owners; document the current baseline.	Purpose statement, role assignment, workflow and baseline record.
Days 16 to 30	Classify the mechanism; map data and supplier boundaries; define reference evidence, tolerances, subgroups and excluded uses.	Task and technology classification dossier.
Days 31 to 60	Run a controlled comparison using public, synthetic, de-identified or formally approved data; record all verification and correction.	Evaluation report, uncertainty ledger and quality-adjusted time.
Days 61 to 75	Test failure scenarios, stopping rules, fallback, permissions, change control and incident escalation.	Control test and recovery record.
Days 76 to 90	Hold an independent challenge review and issue permit, redesign, reject or escalate decision with expiry and monitoring.	Signed decision record and institutional learning note.

Minimum governance instruments

Instrument	Purpose
Use case register	Prevents a bounded experiment from becoming undeclared general approval.
Evidence record	Preserves versions, sources, evaluation, corrections, exceptions and approvals.
Failure and incident register	Connects technical events to statistical products, affected groups and containment.
Stopping and restart protocol	Names triggers, immediate action, fallback, investigation, approval and restart evidence.
Supplier change control	Closes the production gate when a material model, corpus, configuration or service change is not validated.
Review calendar	Forces reconsideration after time, drift, classification revision, source change or material incident.

Executive decision checklist

A proposal is ready for a bounded decision only when leadership can answer every mandatory question with evidence.

Decision test	Yes only when
Purpose	The official purpose, user, population, consequence and excluded decisions are explicit.
Baseline	The current method and simpler alternatives are measured comparably.
Mechanism	The institution can state what is encoded, learned, retrieved, generated or optimized.
Evidence	Independent evaluation, uncertainty, tolerances and subgroup/domain results support the claim.
Quality	Effects on relevance, accuracy, timeliness, coherence, accessibility and reproducibility are assessed.
Confidentiality	Data flows, supplier access, logs, retention, deletion and incident response are approved.
Human control	Review is substantive, resourced and able to correct, stop and recover.
Authority	Named officials retain methodological, operational and release decisions.
Monitoring	Triggers, reference evidence, fallback, expiry and review date are defined.
Decision record	Permit, redesign, reject or escalate is conditional, traceable and clear about residual uncertainty.

Controlled references

- United Nations General Assembly. Fundamental Principles of Official Statistics. Resolution 68/261, 2014.
- United Nations Statistics Division. United Nations National Quality Assurance Frameworks Manual for Official Statistics, 2019.
- UNECE. Machine Learning for Official Statistics, 2022.
- UNECE. Responsible AI for Official Statistics Framework, 2025.
- NIST. Artificial Intelligence Risk Management Framework 1.0, 2023.
- OECD. Explanatory Memorandum on the Updated OECD Definition of an AI System, 2024.
- SRAI. OS-A02-A01 AI Concepts for Official Statisticians v0.2, accepted 26 September 2026.

This brief supports leadership decisions; it does not replace jurisdiction-specific law, institutional security controls, statistical methodology or the authority of designated officials.